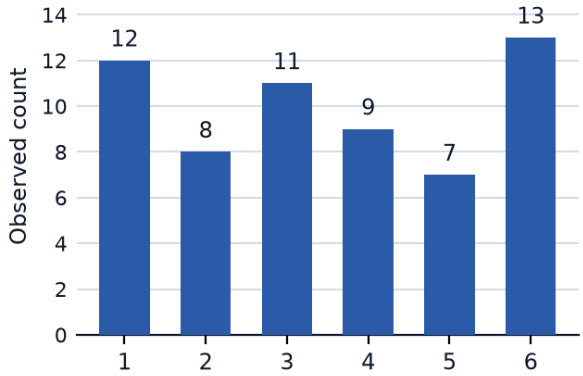
Chi-Square Tests & Regression Inference

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

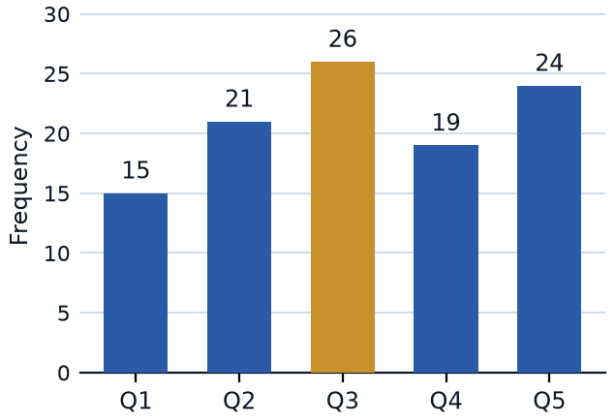
1 [Foundational] • Target: Chi-square goodness-of-fit test

A die is rolled 60 times, expecting each face to appear equally often (expected count 10 per face). The observed counts are $12, 8, 11, 9, 7, 13$. Compute the chi-square test statistic using $\chi^2 = \sum \frac{(O-E)^2}{E}$.



2 [Foundational] • Target: Chi-square goodness-of-fit test

State the null and alternative hypotheses for the goodness-of-fit test in the die-rolling scenario above.



3 [Foundational] • Target: Chi-square test for independence

A two-way table examines whether preferred learning style (visual, auditory, kinesthetic) is associated with academic major (STEM, non-STEM). State the null and alternative hypotheses for a chi-square test of independence in this context.

4 [Foundational] • Target: Chi-square test for independence

In a chi-square test of independence with a 3×2 table (3 learning styles, 2 major categories), find the degrees of freedom, using $df = (\text{rows} - 1)(\text{columns} - 1)$.

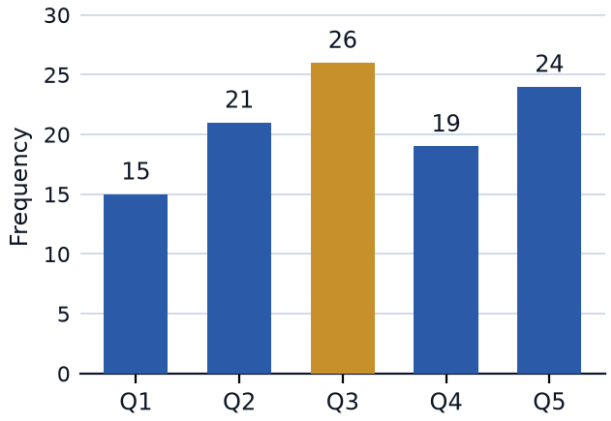
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5 [Foundational] • Target: Interpreting chi-square results

A chi-square test of independence produces a test statistic $\chi^2 = 8.2$ with $df = 2$, and a p-value of 0.017. State a conclusion at the $\alpha = 0.05$ significance level, in context.

6 [Foundational] • Target: Interpreting chi-square results

Explain a condition that must be checked before conducting a chi-square test, related to expected cell counts.



7 [Foundational] • Target: Inference for the slope of a regression line

A regression of exam score on hours studied gives a sample slope $b_1 = 6.2$ with standard error $SE_{b_1} = 1.1$, based on $n = 25$ data points. Construct a 95% confidence interval for the true population slope β_1, using a critical value $t^* \approx 2.069$ (with $df = 23$).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Inference for the slope of a regression line

State the null and alternative hypotheses for testing whether hours studied has a significant linear relationship with exam score, and compute the test statistic using the values from the previous problem.

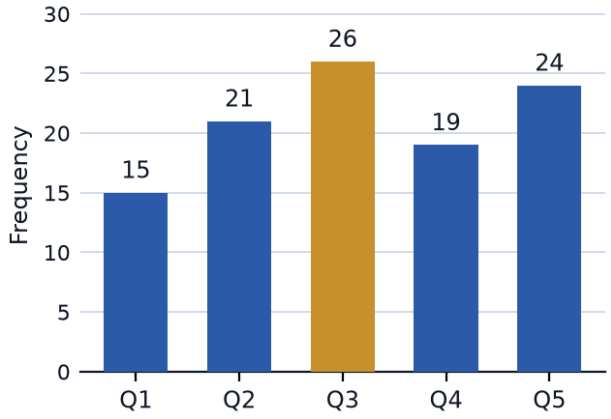
9 [Intermediate] • Target: Conditions for regression inference

State the four conditions (sometimes remembered as LINER: Linear, Independent, Normal, Equal variance, Random) required for valid inference about a regression slope.

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10 **[Intermediate]** • **Target: Putting it together: a full inference scenario**

A study finds a chi-square test statistic of $\chi^2 = 3.1$ with $df = 1$, and a critical value of $\chi^2_{0.05} = 3.84$ at the $\alpha = 0.05$ level. State the conclusion, and explain how this differs from a p-value based approach.



11 **[Intermediate]** • **Target: Chi-square test for homogeneity**

Explain the difference between a chi-square test of independence and a chi-square test of homogeneity, even though both use the same formula and table structure.

12 **[Intermediate]** • **Target: Computing expected counts for a chi-square test**

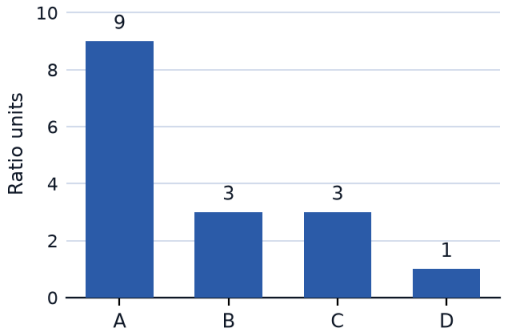
In a two-way table, a row total is 80, a column total is 120, and the grand total is 300. Find the expected count for that cell, using $E = \frac{\text{(row total)(column total)}}{\text{grand total}}$.

13 **[Intermediate]** • **Target: Computing expected counts for a chi-square test**

State the degrees of freedom for a chi-square goodness-of-fit test with 6 categories, using $df = (\text{number of categories}) - 1$.

14 **[Intermediate]** • **Target: Computing expected counts for a chi-square test**

A genetics experiment expects offspring to appear in a 9:3:3:1 ratio across four phenotype categories, based on 160 total offspring observed. Find the expected count for each category, and briefly describe what the chi-square goodness-of-fit test would examine using these expected counts.



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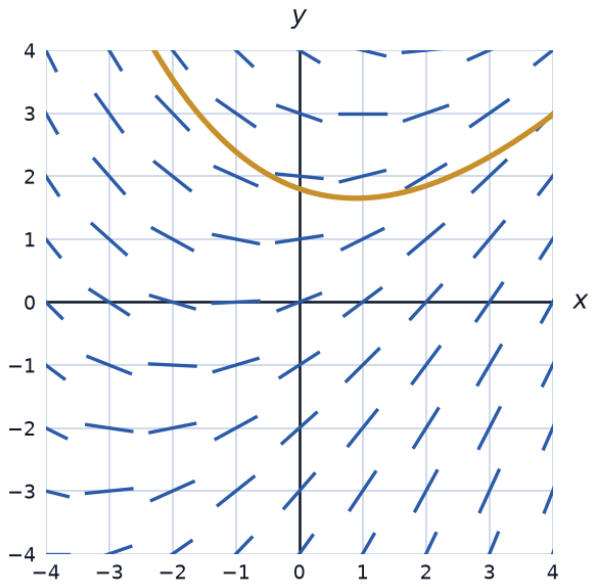
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

A study examines the relationship between hours of sleep per night (x) and reaction time in milliseconds (y) for 30 participants, finding a regression slope $b_1 = -8.5$ with standard error $SE_{b_1} = 2.3$. State the hypotheses for testing whether more sleep is associated with a significant decrease in reaction time (a one-sided test), compute the test statistic, and interpret the sign of the slope in context.

16 **[Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A university surveys 240 students, recording both their preferred study location (library, dorm, coffee shop) and their class year (underclassman, upperclassman), to test whether study location preference is independent of class year. The resulting chi-square test statistic is $\chi^2 = 11.4$ with $df = 2$, and the corresponding p-value is 0.0033. State the conclusion at $\alpha = 0.01$, and explain, in the context of this scenario, what rejecting or failing to reject H_0 would mean for the university's understanding of its students.



17 **[Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions**

A researcher studies the relationship between advertising spending (in thousands of dollars, x) and monthly sales (in thousands of dollars, y) for 22 small businesses, obtaining a regression slope $b_1 = 3.2$ with standard error $SE_{b_1} = 0.85$. Construct a 95% confidence interval for the true population slope β_1 using a critical value $t^* \approx 2.086$ (with $df = 20$), and interpret the interval in context, including whether it suggests advertising spending has a statistically significant positive association with sales.

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18 [Challenging] • Target: Putting it together: a full inference scenario • Cross-Lesson: 3.03 The Normal Distribution

Eight patients with chronic insomnia try a new relaxation therapy. Their sleep duration (hours/night) is measured before and after 4 weeks of therapy: Patient differences $d = \text{After} - \text{Before}$ yield $\bar{x}_d = 1.25$ hours and $s_d = 0.80$ hours. Assuming normality of differences, test whether the therapy significantly increased sleep at $\alpha = 0.01$.

19 [Challenging] • Target: Chi-square test for homogeneity • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

In the tomato experiment in Problem 7, the agronomist has two distinct tomato varieties: Roma and Beefsteak. (a) Explain why a randomized block design is superior to a completely randomized design. (b) Outline the blocking protocol.

20 [Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A researcher is investigating the relationship between study habits and exam performance. They categorize students into two groups: 'Regular Study' and 'Cramming'. They also categorize exam performance into 'Pass' and 'Fail'. The data collected is presented in a two-way table. If the table has 2 rows representing study habits and 3 columns representing exam performance categories, what is the total number of cells in this two-way table?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Chi-square goodness-of-fit test

A survey expects 4 categories to appear equally often, with expected count $E = 15$ per category. The observed counts are $17, 21, 13, 9$. Compute the chi-square test statistic using $\chi^2 = \sum (O - E)^2 / E$.

a. 5.33

b. 4.33

c. 5.00

d. 6.33

22 [Foundational] • Target: Chi-square test for independence

A survey expects 4 categories to appear equally often, with expected count $E = 20$ per category. The observed counts are $24, 14, 18, 24$. Compute the chi-square test statistic using $\chi^2 = \sum (O - E)^2 / E$.

a. 4.60

b. 3.60

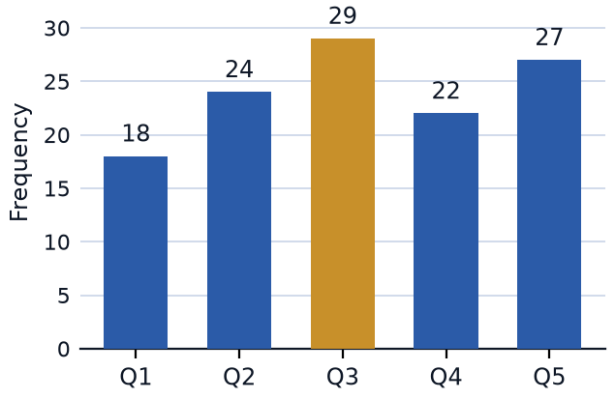
c. 2.60

d. 3.43

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23 [Foundational] • Target: Interpreting chi-square results

A survey expects 4 categories to appear equally often, with expected count $E = 11$ per category. The observed counts are $6, 16, 9, 13$. Compute the chi-square test statistic using $\chi^2 = \sum(O - E)^2/E$.



a. 4.83

b. 6.27

c. 4.27

d. 5.27

24 [Foundational] • Target: Inference for the slope of a regression line

A survey expects 4 categories to appear equally often, with expected count $E = 15$ per category. The observed counts are $10, 18, 12, 20$. Compute the chi-square test statistic using $\chi^2 = \sum(O - E)^2/E$.

a. 5.53

b. 4.53

c. 4.25

d. 3.53

25 [Foundational] • Target: Conditions for regression inference

A survey expects 4 categories to appear equally often, with expected count $E = 24$ per category. The observed counts are $27, 26, 25, 18$. Compute the chi-square test statistic using $\chi^2 = \sum(O - E)^2/E$.

a. 3.08

b. 2.00

c. 2.08

d. 1.08

26 [Foundational] • Target: Putting it together: a full inference scenario

A survey expects 4 categories to appear equally often, with expected count $E = 15$ per category. The observed counts are $21, 9, 16, 14$. Compute the chi-square test statistic using $\chi^2 = \sum(O - E)^2/E$.

a. 3.93

b. 4.93

c. 5.93

d. 4.62

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27 [Foundational] • Target: Chi-square test for homogeneity

A survey expects 4 categories to appear equally often, with expected count $E = 14$ per category. The observed counts are $15, 18, 11, 12$. Compute the chi-square test statistic using $\chi^2 = \sum(O - E)^2/E$.

a. 1.14

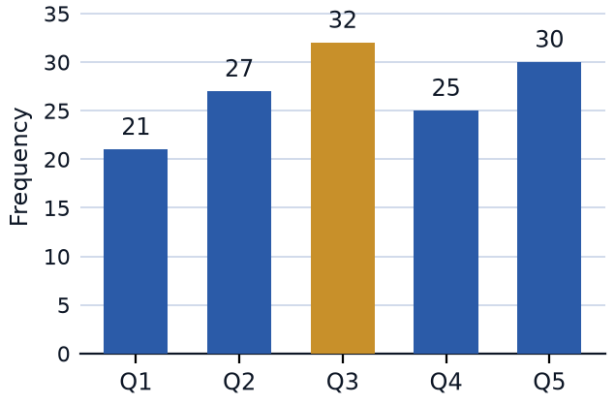
b. 3.14

c. 2.00

d. 2.14

28 [Foundational] • Target: Computing expected counts for a chi-square test

In a two-way table, a row total is 87, a column total is 46, and the grand total is 200. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.



a. 23.01

b. 112.99

c. 20.01

d. 17.01

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Chi-Square Goodness-of-Fit Test

In a two-way table, a row total is 266, a column total is 99, and the grand total is 400. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 68.83

b. 65.83

c. 62.84

d. 299.17

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30 [Intermediate] • Target: Chi-Square Test of Independence vs. Homogeneity

In a two-way table, a row total is 250, a column total is 314, and the grand total is 500. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 160.00

b. 157.00

c. 407.00

d. 154.00

31 [Intermediate] • Target: Inference for Linear Regression Slope

In a two-way table, a row total is 103, a column total is 50, and the grand total is 150. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 118.67

b. 31.33

c. 37.33

d. 34.33

32 [Intermediate] • Target: Regression Model Conditions (LINER)

In a two-way table, a row total is 154, a column total is 295, and the grand total is 500. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 358.14

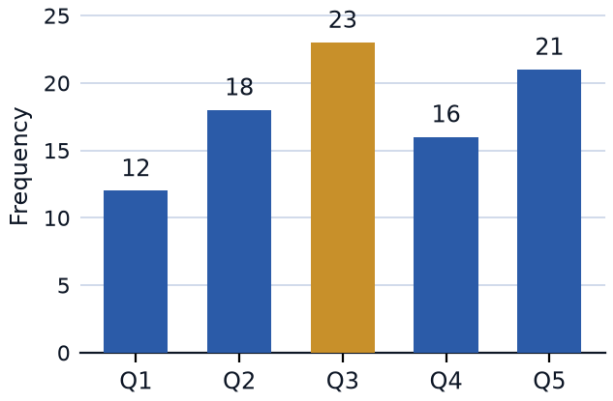
b. 87.86

c. 90.86

d. 93.86

33 [Intermediate] • Target: Chi-square goodness-of-fit test

In a two-way table, a row total is 96, a column total is 134, and the grand total is 200. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.



a. 64.32

b. 61.32

c. 165.68

d. 67.32

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34 [Intermediate] • Target: Chi-square test for independence

In a two-way table, a row total is 65, a column total is 80, and the grand total is 250. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 23.80

b. 17.80

c. 124.20

d. 20.80

35 [Intermediate] • Target: Interpreting chi-square results

A regression gives a sample slope $b_1 = -5.2$ with standard error $SE_{b_1} = 1.4$, based on data with $df = 33$. Construct a 95% confidence interval for the true population slope β_1, using a critical value $t^* \approx 2.03$.

a. $(-8.04, -2.86)$

b. $(-9.04, -1.36)$

c. $(-7.54, -2.36)$

d. $(-8.04, -2.36)$

36 [Intermediate] • Target: Inference for the slope of a regression line

A regression gives a sample slope $b_1 = 3.1$ with standard error $SE_{b_1} = 0.6$, based on data with $df = 15$. Construct a 95% confidence interval for the true population slope β_1, using a critical value $t^* \approx 2.131$.

a. $(1.82, 3.88)$

b. $(0.82, 5.38)$

c. $(2.32, 4.38)$

d. $(1.82, 4.38)$

37 [Intermediate] • Target: Conditions for regression inference

A regression gives a sample slope $b_1 = 6.8$ with standard error $SE_{b_1} = 1.25$. Compute the test statistic for testing $H_0 : \beta_1 = 0$, using $t = b_1 / SE_{b_1}$, to two decimal places.

a. 6.44

b. -5.44

c. 4.44

d. 5.44

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Putting it together: a full inference scenario • Cross-Lesson: 3.03 The Normal Distribution

A regression gives a sample slope $b_1 = 6.2$ with standard error $SE_{b_1} = 1.25$. Compute the test statistic for testing $H_0 : \beta_1 = 0$, using $t = b_1/SE_{b_1}$, to two decimal places.

a. -4.96

b. 3.96

c. 4.96

d. 5.96

39 [Challenging] • Target: Chi-square test for homogeneity • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A chi-square test produces a test statistic $\chi^2 = 13.7$ with $df = 5$, and a critical value of $\chi^2_{0.05} = 11.07$ at $\alpha = 0.05$. State the conclusion, comparing the test statistic directly to the critical value.

a. The test cannot be completed without a p-value

b. Reject H_0; there is convincing evidence of an association

c. Fail to reject H_0; there is not convincing evidence of an association

d. Accept H_0 as definitely true

40 [Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A chi-square test produces a test statistic $\chi^2 = 1.2$ with $df = 6$, and a critical value of $\chi^2_{0.05} = 12.59$ at $\alpha = 0.05$. State the conclusion, comparing the test statistic directly to the critical value.

a. The test cannot be completed without a p-value

b. Accept H_0 as definitely true

c. Fail to reject H_0; there is not convincing evidence of an association

d. Reject H_0; there is convincing evidence of an association

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41 [Challenging] • Target: Chi-Square Goodness-of-Fit Test • Cross-Lesson: 1.01 Measures of Center & Spread

What is the formula for the Chi-Square test statistic?

a. $\chi^2 = \sum \frac{(O-E)^2}{E}$

b. $\chi^2 = \sum \frac{(O-E)}{E}$

c. $\chi^2 = \sum \frac{(O-E)^2}{O}$

d. $\chi^2 = \frac{(\bar{x}-\mu)^2}{s^2}$

42 [Challenging] • Target: Chi-Square Goodness-of-Fit Test • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

What condition must be satisfied regarding expected counts for a Chi-Square test to be valid?

a. All expected cell counts must be at least 5

b. All observed cell counts must be at least 30

c. The sample size must be exactly 100

d. The categories must be normally distributed

43 [Challenging] • Target: Chi-Square Test of Independence vs. Homogeneity • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

What are the degrees of freedom for a Chi-Square test on a 3×4 contingency table?

a. 6

b. 12

c. 7

d. 11

44 [Challenging] • Target: Inference for Linear Regression Slope • Cross-Lesson: 2.01 Sampling Methods & Bias

Computer regression output shows: Predictor Coef SE Coef T P; Constant 14.2 2.1 6.76 0.000; Slope 3.45 0.69 5.00 0.0001. For $n = 18$ observations, what is the t-statistic for the slope?

a. 5.00

b. 6.76

c. 3.45

d. 0.69

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45 [Challenging] • Target: Regression Model Conditions (LINER) • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

What does the acronym LINER stand for regarding the conditions for regression inference?

a. Linear, Independent, Normal y for each x, Equal variance, Random sampling

b. Least squares, Intercept, Normal, Errors, Residuals

c. Linear, Integrated, Numerical, Estimated, Robust

d. Limit, Interval, Normal, Expected, Ratio

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